EXAM SHIELD

Python Question Bank

Core syntax, data types and idioms of Python 3.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 20 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Python

Core syntax, data types and idioms of Python 3.

1. Which of these Python types is immutable?

A. dict
B. tuple
C. set
D. list

Answer: B

2. Which keyword defines a function in Python?

A. def
B. func
C. fn
D. function

Answer: A

3. What does len("hello") return?

A. 6
B. 4
C. An error
D. 5

Answer: D

4. What is the result of the expression 7 // 2 in Python 3?

A. 1
B. 3.5
C. 3
D. 4

Answer: C

5. Which line is a valid dictionary comprehension?

A. [k: v for k, v in pairs]
B. (k: v for k, v in pairs)
C. {k: v for k, v in pairs}
D. dict[k for k in pairs]

Answer: C

6. What does a with open(path) as fh: block guarantee?

A. The file is closed when the block exits
B. The file is locked for other processes
C. The file is deleted afterwards
D. The file is opened read-only

Answer: A

7. Which tool installs third-party Python packages?

A. npm
B. gem
C. composer
D. pip

Answer: D

8. What does print(type([])) display?

A. <class 'list'>
B. <class 'dict'>
C. <class 'array'>
D. <class 'tuple'>

Answer: A

9. Python handles exceptions with a try block followed by one or more _____ blocks, and code in a _____ block always runs. [2 marks]

Answer: except | finally

10. The special method that initialises a new object of a Python class is named _____. [1 mark]

Answer: __init__

11. An isolated per-project Python environment is created with the standard library module _____. [1 mark]

Answer: venv

12. Explain the Global Interpreter Lock (GIL) and its effect on multi-threaded, CPU-bound Python code. [4 marks]

13. Compare a list with a generator. When is a generator the better choice? [4 marks]